Spokane Area Fires: Facebook Data Situation Report

NSF AI-SDM and Meta AI for Good

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Key take-aways

- Key Outflow & Influx Zones (as of Aug. 12 evening): Population outflow is highest near the northeastern section of the Old Trails Fire, while incoming movements are primarily directed toward Airway Heights and central Spokane.
- Evolution of Influx Hubs: When data first became available on Aug. 3, urban Spokane (likely around the Convention Center shelter) saw the largest influx of people. By Aug. 12, Airway Heights became the primary destination for incoming populations.
- Activity Trends Over Time: Movement volume peaked on Aug. 3 and has since tempered as of Aug. 12, indicating a gradual stabilization, though population movement remains elevated relative to the 90-day baseline.
- Population Presence Within Fire Zones (Aug. 12): Despite significant evacuation from the northeast Old Trails Fire area compared to baseline, population density inside parts of the perimeter remains high. Conversely, the Autumn Lane fire zone shows fewer people.

1. Per-tile user population: Difference in crisis window vs. pre-crisis baseline

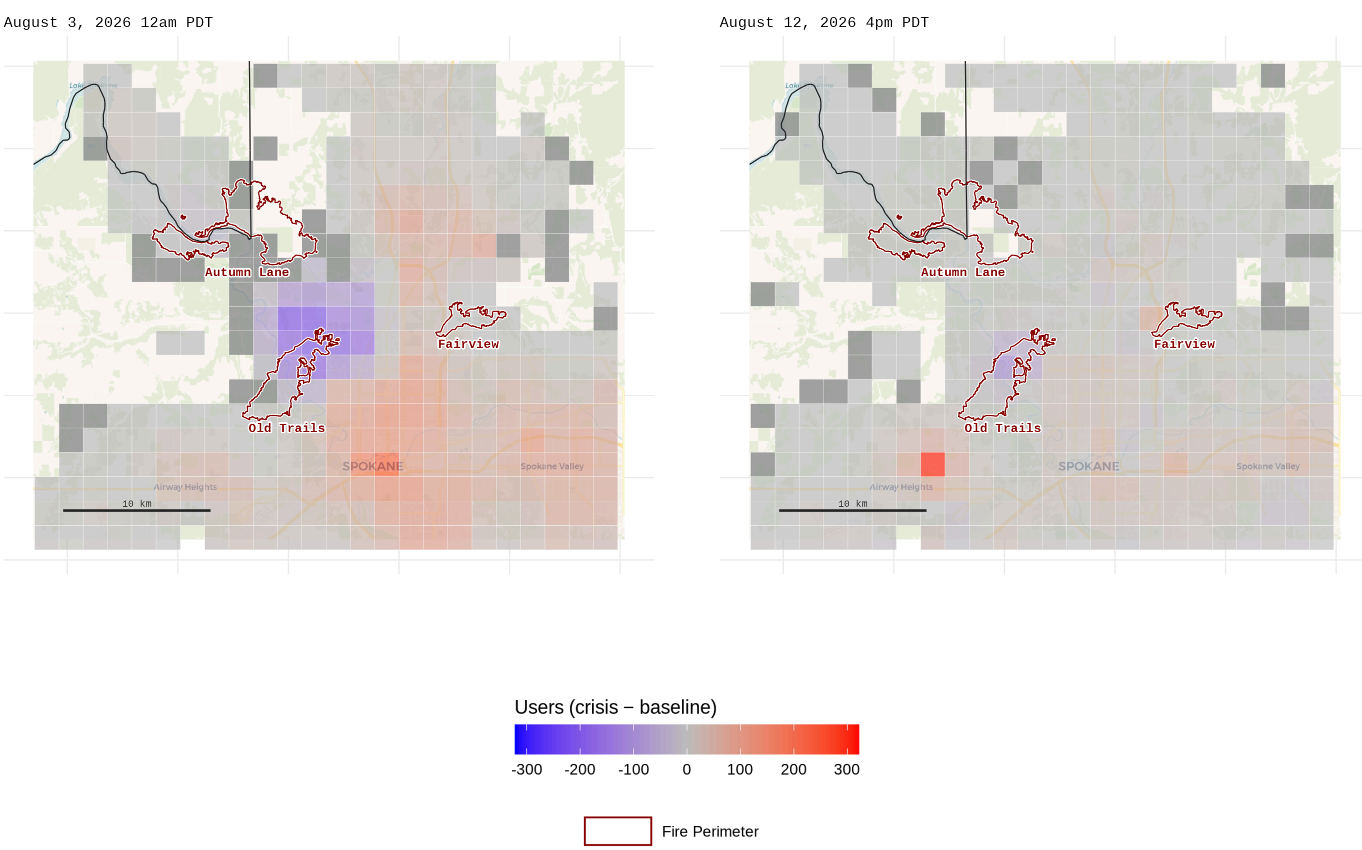

Figure 1: Difference (crisis − baseline) per tile; left = first window, right = latest. Shared symmetric scale + one combined legend. Blue = fewer than baseline (exodus), red = more (influx), grey = NA.

The difference maps below plot (crisis − baseline) per tile, with fire perimeters in dark red (WFIGS *current* feed, cached August 17, 2026 10:05 AM PDT). The strongest exodus (blue) hugs the northeast fire perimeter nearest the metro region. The strongest influx (red) sits in Spokane's urban core on Aug 3 (possibly given the Convention Center shelter), and later in the in the Airway Heights region on Aug 12.

On Aug. 3 (the first day of available data), people movements were more acute compared to Aug. 12. This indicates that people movements are starting to wane as the fire progresses - but not yet reverting to pre-fire levels (vs. a prior 90-day baseline).

2. Raw user counts

Raw users per tile given latest data (red = more, blue = fewer) indicate that the metro region is the most transited, even when close to fire perimeters. Red stays concentrated in Spokane's urban core. Almost no users are near the Autumn Lane fire, while some remain near the Fairview Fire as well as the northeast corner of the Old Trails Fire. Both panels share one scale and one combined legend.

Note that despite the general exodus from the northeast corner of the Old Trails Fire (relative to a prior 90-day baseline of user count), raw user counts are at around 170-300 users across several 1.6-km square tiles. The Autumn Lane fire has far fewer users, less than 50 per 1.6-km square tile.

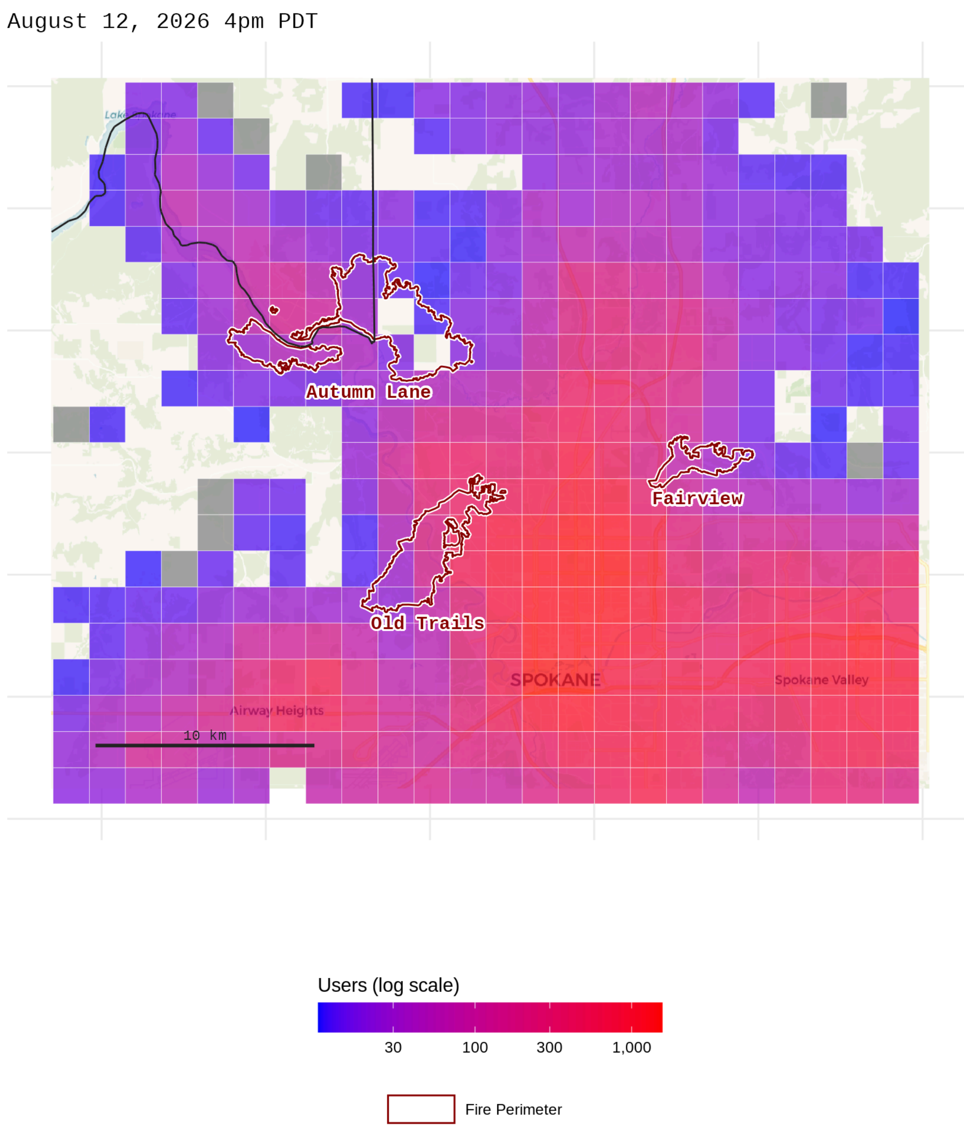

Figure 2: Raw crisis user counts per tile; right = latest. Shared scale + one combined legend. Red = more users, blue = fewer. Shows where users ARE, not where change occurred.

Appendix

Incident	Spokane Area Fires: Old Trails, Autumn Lane, Fairview
Total size	~10,332 acres (Old Trails 3,556; Autumn Lane 5,784; Fairview 992)
Containment	Old Trails 43%, Autumn Lane 47%, Fairview 31% (Aug 10)
Origin	Aug 1, 2026 15:00 PDT; cause under investigation
Location	Spokane & Stevens Counties, WA, USA
Evacuation	Levels 1–3 (Be Ready / Be Set / Go Now); see spokane911.com for latest
Data available	August 3, 2026 12am PDT to August 12, 2026 4pm PDT; fire ongoing as of Aug 13

This report summarizes Meta Data for Good population and movement signals for the days of available data (Aug 3–12) to help the Incident Management Team (IMT) see **where** anomalous population change is concentrated and **whether** it is consistent with the active evacuation orders. It is a decision-support supplement, not a headcount: every figure reflects Facebook users with Location Services on, not the whole population (see Data caveats).

Data description

We use Meta's **Facebook Movement During Crisis** and **Facebook Population During Crisis** datasets, which display location activity from Facebook users (with **Location Services on**) used to estimate how population shifts during a crisis. The data represent **Facebook app users**, not everyone in the area — henceforth "users." The datasets compare user behavior geographically near the incident with pre-crisis behavior. There are two granularities: fixed **8-hour windows** (starting 00:00 / 08:00 / 16:00 Pacific) and **Bing tiles** (level 14, ~2.4 km or ~1.5 mi squares). For privacy, a window is shown as NA / grey when either the crisis **or** the baseline count is under 10 users. Grey indicates "one side was below threshold" (not "no change" or "fewer than 10 people total"). Two signals are used: comparison to a **pre-crisis baseline** for the same time-of-day / day-of-week (a 90-day baseline for movement and population datasets) and **raw user counts** are determined at window *T*.

As of this report the latest **population** window is August 12, 2026 4pm PDT and the latest **movement** window is August 12, 2026 4pm PDT. Population maps (§1–§2) therefore show the 8-hour window ending August 12, 2026 4pm PDT.

> ⚠ Data caveats
>
> - **Sample bias.** this data covers only Facebook users with Location Services turned on.
> - **Data intervals.** 8-hour windows, up to ~8 h delay.
> - **Tile size.** ~2.4 km or ~1.5 mi, neighborhood-scale, not address-level.
> - **Privacy controls.** A window is NA / grey when the crisis or baseline count is <10 (both <10 means the row is dropped).
> - **Baselines.** Movement and population data use a 90-day baseline before the crisis.
> - **Evacuation vs connectivity.** A tile emptying could mean people left **or** power/cell dropped.
> - **Who is under-counted.** The signal skews toward mobile, connected, vehicle-owning residents and may under-capture access-and-functional-needs populations, such as people with mobility limits, no vehicle, or no transit option.
> - **Users =/= people =/= vehicles.** Counts are a fraction of residents, and residents may map to more vehicles still (according to AAR, most local households evacuate with two). These figures are not a road-traffic count.

1. Per-tile user population: crisis window vs. pre-crisis baseline

The table compares users per tile in the first available crisis window (August 3, 2026 12am PDT) to the pre-crisis baseline for the same time-of-day / day-of-week 90 days earlier. *Most* tiles are near baseline, but a few show large **differences**.

Table 1

Number of Users	Min	Average	Median	Max
Crisis (August 3, 2026 12am PDT)	10	117.3	26.8	1186.2
Baseline (90-day, same window)	10	115.0	26.4	1085.6

Out of 817 tiles, 17 had a difference ≤ −50 (people leaving, ≈ 38 sq mi) and 68 had ≥ 50 (people entering, ≈ 153 sq mi).

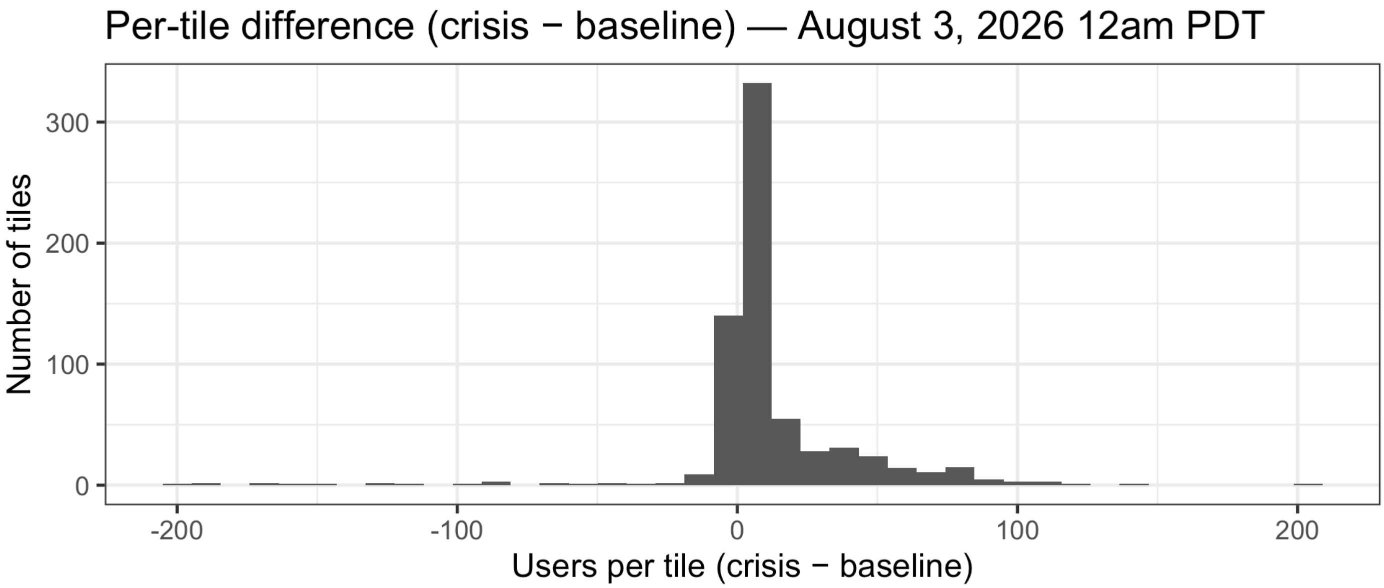

Figure 3: Distribution of per-tile user count differences (crisis vs. 90-day baseline), first window.

1a. Standardized change (z-score)

The difference maps use data-driven color limits, so one large-influx tile can wash the ±50 signal to pale grey. The z-score is Meta's recommended metric, and it captures each tile's change in standard deviations vs its own baseline, clipped to ±4 — so anomalies stand out. Blue = fewer users than normal (exodus); red = more users (influx); grey = no data. Because it is measured against each tile's *own* pre-fire baseline, the z-score separates *left because they evacuated* from *empty because it is rural* — which the raw-count map (§2) cannot — so it is the map we use to orient this report to the official evacuation areas.

Orienting the maps to the evacuation areas

The two panels below place the latest-window z-score (zoomed to the Spokane metro and the nearest fire, where the evacuation signal is concentrated) beside a screenshot of the county's official evacuation areas (Spokane County / SREC *Evacuation Address Lookup*). Both panels show the **same fires** — the dark-red perimeters on the right and the **Level 3 (Go Now)** areas on the left wrap the same fire — so a reader can line the two up by the fire shapes and then carry that orientation to every other map in this report.

Note: the left panel is a static, non-georeferenced screenshot, so the alignment is approximate and by eye.

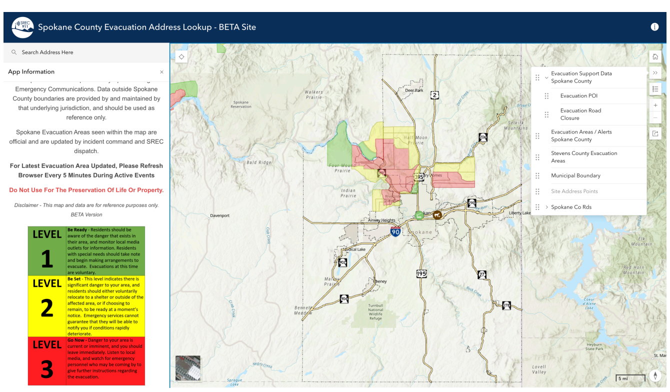

Figure 1: **Official evacuation areas** — Spokane County / SREC *Evacuation Address Lookup* (BETA), captured Aug 9, 2026 ~00:10 PT. Green = Level 1 (Be Ready), yellow = Level 2 (Be Set), red = Level 3 (Go Now). Per SREC the map is for reference only; check webpage for latest evacuation zones.

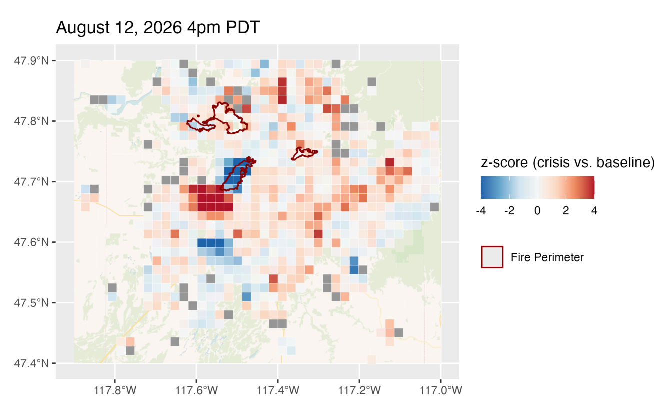

Figure 2: **Population anomaly (z-score), latest data window** — blue = below normal (exodus), red = above (influx), grey = NA; fire perimeters in dark red. Same fires as the left panel.

Interpretation. Where the compact **blue (exodus)** cluster on the right falls inside a **red Level 3 (Go Now)** area on the left, the movement signal is consistent with the order. A fire-adjacent tile that is at or above normal (grey-to-red) *inside* a red zone would instead flag **possible residents remaining** — worth checking against ground reports, since the city's 2026 wildfire evacuation-drill after-action review (AAR) found a meaningful share of residents do not leave immediately even after a Go Now alert. *Timing:* the z-score panel is the 8-hour window ending August 12, 2026 4pm PDT.

Now the first vs. latest comparison, on the same fixed ±4 scale with one combined legend:

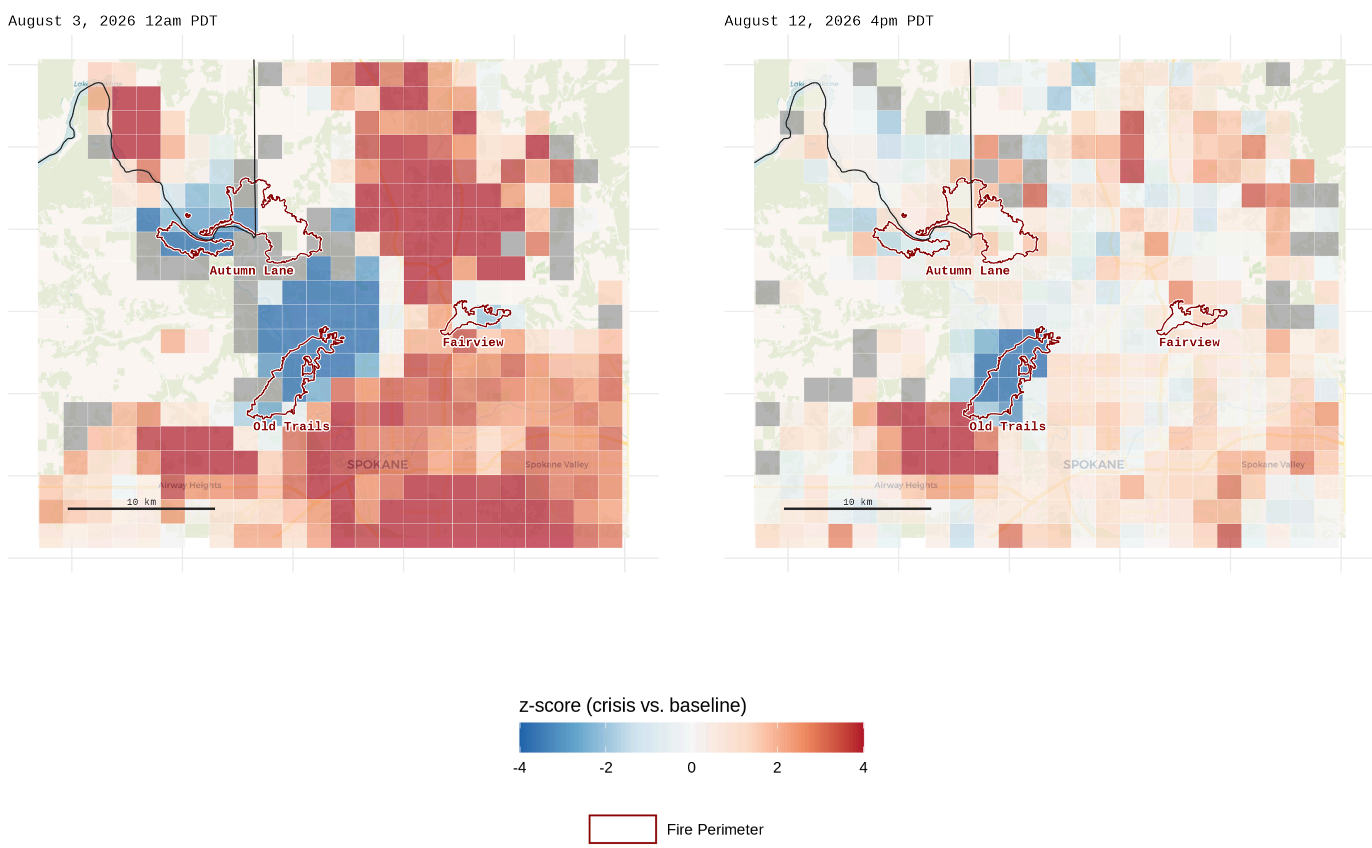

Figure 4: Standardized change (z-score, RdBu, clipped ±4); left = first window, right = latest. The ±4 scale is fixed and shared, so the panels are directly comparable. Blue = exodus, red = influx, grey = NA (not 'no change').